

Capstone 3 Final Report: NYC Taxi Trip Duration Analysis

1. Problem Statement

The goal of this project is to analyze NYC taxi trip data and predict trip duration using machine learning models. Understanding the factors associated with taxi travel time can help improve transportation planning, routing efficiency, and overall passenger experience within urban transportation systems.

2. Data & Preprocessing

The NYC Yellow Taxi dataset was cleaned and prepared for modeling. Datetime columns were converted into proper datetime format, and trip duration was calculated using pickup and dropoff timestamps. Invalid and extreme trip durations were removed to reduce the influence of unrealistic outliers. Additional time-based features, such as pickup hour and pickup day, were created to capture temporal travel patterns.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify relationships between trip characteristics and taxi trip duration. The analysis showed that trip distance was strongly associated with trip duration, while passenger count had relatively weak correlation. Time-based and geographic features also contributed to variability in travel time.

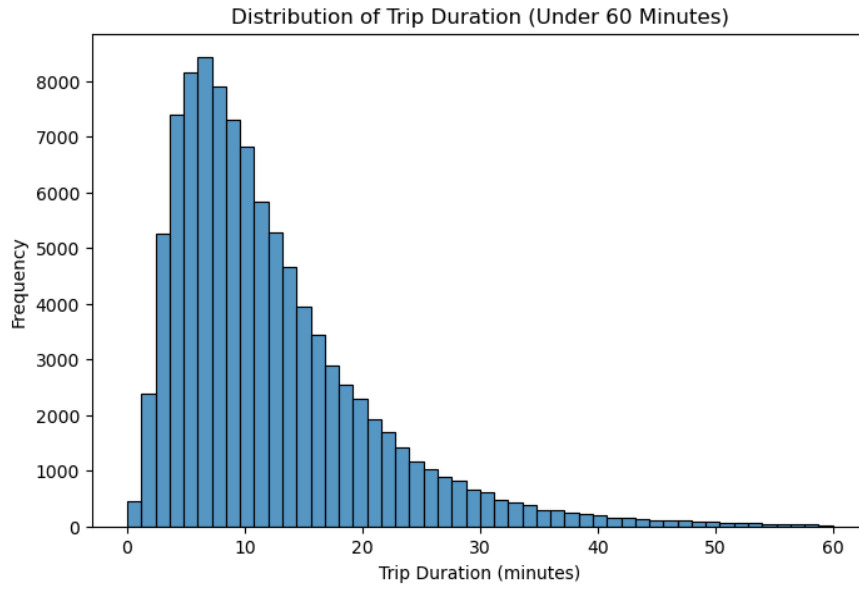


Figure 1: Distribution of Trip Duration

The distribution of trip duration was heavily right-skewed, with most taxi trips lasting a relatively short amount of time and a small number of extremely long trips acting as outliers. This indicated the need to remove unrealistic values prior to modeling.

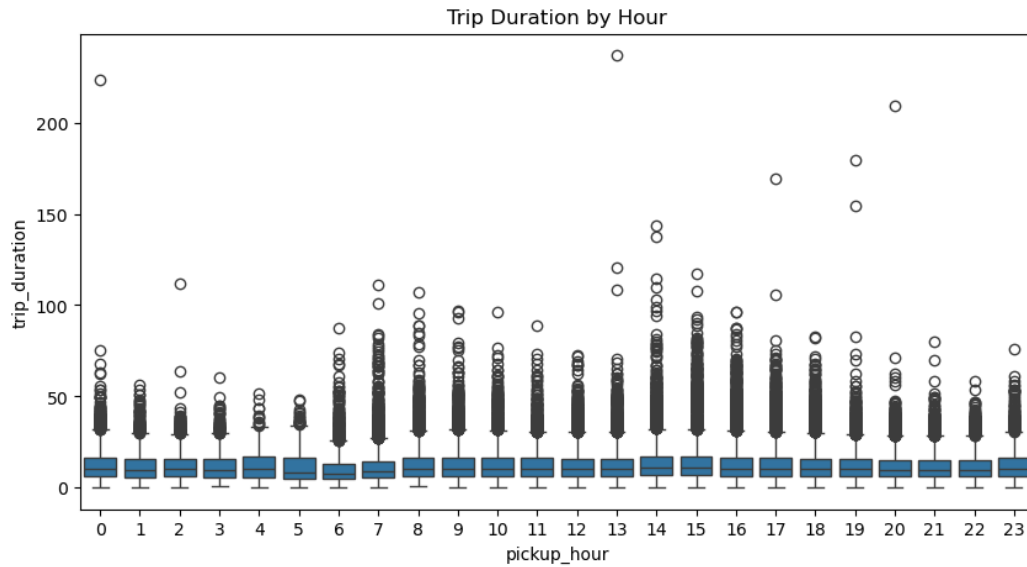


Figure 2: Average Trip Duration by Pickup Hour

Average trip duration varied throughout the day, with longer trips often occurring during high-traffic commuting periods. This suggests that traffic congestion and time-of-day patterns influence travel time across New York City.

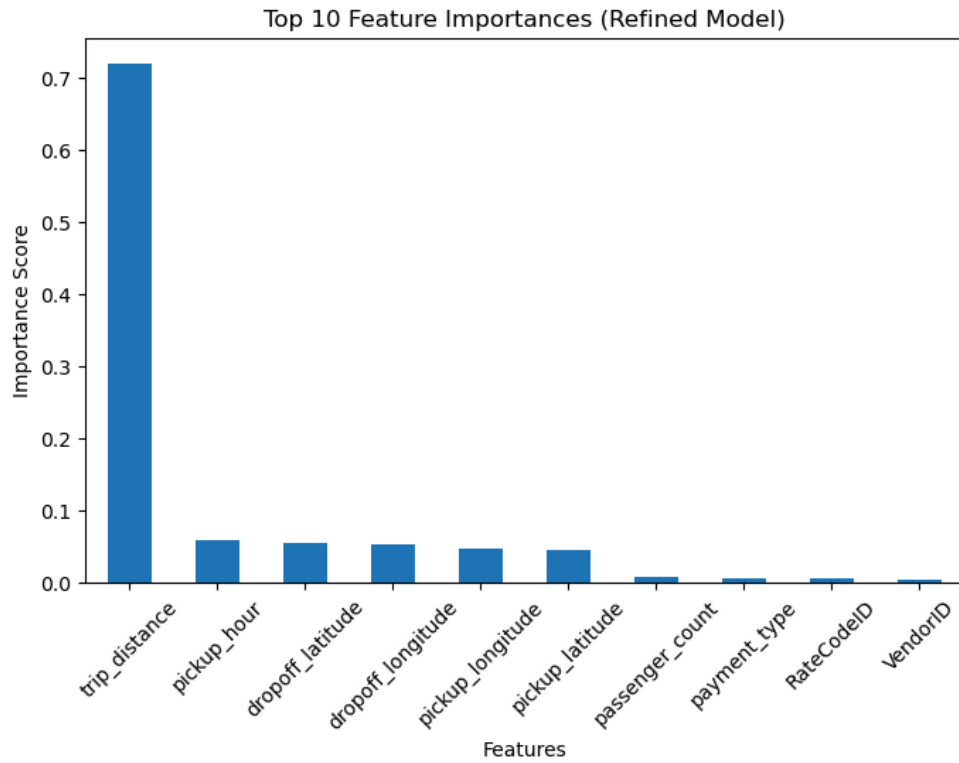


Figure 3: Feature Importance – Refined Random Forest Model

Trip distance emerged as the strongest independent predictor of trip duration in the refined Random Forest model. Geographic and time-based features also contributed to prediction performance, suggesting that both route length and urban travel conditions influence taxi travel time.

4. Modeling Approach

The dataset was split into training and testing sets to evaluate performance on unseen data. Feature scaling was applied for Linear Regression and Ridge Regression models using `StandardScaler`, while Random Forest models used unscaled data due to the tree-based structure of the algorithm.

Three primary models were trained:

- Linear Regression as a baseline model
- Ridge Regression to reduce potential overfitting and stabilize coefficients
- Random Forest Regression to capture non-linear relationships within the dataset

An additional refined Random Forest model was trained after removing fare-related variables to better evaluate the independent predictors of trip duration.

5. Model Results

Linear Regression and Ridge Regression produced similar results, each achieving an R^2 score of approximately 0.71 and an RMSE of approximately 5.0. The original Random Forest model achieved the strongest predictive performance overall, with an R^2 score of approximately 0.88 and an RMSE of approximately 3.29.

Feature importance analysis revealed that fare-related variables dominated the original Random Forest model, suggesting that these features contained substantial information closely tied to completed trip characteristics. To further investigate this relationship, a refined Random Forest model was trained after removing fare-related variables.

The refined Random Forest model achieved an R^2 score of approximately 0.74 and an RMSE of approximately 4.80. Although predictive performance decreased slightly, the refined model provided a more realistic interpretation of the independent factors influencing trip duration.

6. Final Model Selection

The refined Random Forest model was selected as the final model because it provided a more interpretable and realistic representation of the independent variables associated with taxi trip duration. While the original Random Forest model achieved stronger predictive performance, the refined model reduced reliance on fare-related variables closely tied to completed trip outcomes.

7. Recommendations

- Use trip distance and time-based patterns to better estimate expected travel time.
- Incorporate geographic travel patterns into transportation planning and routing analysis.
- Further investigate traffic and congestion trends during high-volume commuting periods.
- Continue refining feature selection to improve interpretability and reduce leakage-like behavior.

8. Conclusion / Future Work

This project demonstrated that trip duration in NYC taxi transportation is influenced by multiple interacting factors, including trip distance, pickup timing, and geographic location. While Random Forest Regression produced the strongest predictive performance overall, the project also highlighted the importance of feature selection and model interpretation when evaluating predictive models.

Future improvements could include incorporating weather data, traffic information, holiday indicators, and advanced boosting algorithms such as XGBoost to further improve prediction performance and model interpretability.